Haotong Luan

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Education

Shenzhen Technology University (SZTU) Sept 2024 – Present
B.Eng. in Automation, Future Technology School; GPA: 83.7/100

Research focus: underactuated grippers; environment-adaptive robotic hands; mechanism design.

Tsinghua University – Shenzhen X-Institute Sept 2025 – Present
P-SRT Program, Joint Training (Underactuated Robotics)

Advisor: Associate Prof. Wenzeng Zhang. Track: X-idea → **P-SRT (current)** → E-SRT → ORIC → SURF.

Publications

Legend: * Corresponding author

Published

Domain-Semantic Subspace Stacking for Financial Fraud Detection
H. Luan, M. Chen, Z. Lin, H. Ding, Y. Sheng*
ICIC 2026 (International Conference on Intelligent Computing). Accepted, first author. 2026

An Underactuated Dual-Slide Adaptive Robotic Hand with Strong Environmental Adaptability
H. Luan, W. Zhang*, T. Yang*
IEEE RAAI 2025 2025

GLINT: An Idle-Stroke Grasp-and-Lift Hand for In-Hand Manipulation
H. Ding, **H. Luan**, T. Yang*, W. Zhang*
IEEE ICCMA 2025 2025

An Idle-Stroke Underactuated Gripper with Independent Double-Parallelogram Linkages for Pinching and Adaptive Grasping
Y. Chen, **H. Luan**, H. Ding, W. Zhang*
IEEE RAAI 2025 2025

Under Review

A Robot Hand with Strong Environmental Adaptability for Grasping, Scooping, and Lifting Based on a Hybrid Parallelogram Mechanism Design
H. Luan, W. Zhang*
IEEE/RSJ IROS 2026 (under review, first author) 2026

SP Hand: An Underactuated Gripper with Mode Switching for Adaptive Pinching and Scooping
H. Luan, W. Zhang*
IEEE/RSJ IROS 2026 (under review, first author) 2026

PlaceRecover: Layout-Invariant Point Cloud Recovery for Robust 3D Perception under LiDAR Placement Shifts
(co-author), **H. Luan**
IEEE/RSJ IROS 2026 (under review) 2026

Patents Pending

- **3 invention patents** filed and under review.
- **1 utility model patent** filed and under review.

Research & Projects

Underactuated Robotic Gripper Research 2025 – Present
Shenzhen X-Institute, Tsinghua University
Designed underactuated grippers using linear-motion linkages, parallelogram mechanisms, and idle-stroke transmissions. Focus: embedding grasping logic into hardware for reliable single-actuator grasping. Led the DS Hand (first-author, RAAI 2025); collaborated on idle-stroke and GLINT gripper designs.

Environment-Adaptive Robotic Hand Design 2025 – Present
Shenzhen X-Institute
Developed hands that exploit contact with tables or walls to switch modes automatically (pinching, scooping, lifting). Designed hybrid parallelogram and slot-guided mechanisms for precise linear fingertip motion and posture-preserving grasping. Two first-author papers submitted to IROS 2026.

Visual-Servoing Control for Robotic-Arm Grasping 2026 – Present
Shanghai AI Laboratory (Remote RA)
Collaborating with Researcher Yuan Gao on visual-servoing control algorithms and end-effector design for robotic-arm grasping. Responsible for the mechanical gripper design mounted on the arm.

Quantum & AI Winter School Jan 2026
Shenzhen Loop Area Institute
Studied generative-model-based quantum variational circuit simulation (MLP / LSTM / Transformer) under Prof. Peng Zhang (Tianjin University). Worked on state fidelity optimization, mixed-state simulation, and physics-constrained training.

Experience

P-SRT Researcher – Shenzhen X-Institute Sept 2025 – Present
Advisor: Associate Prof. Wenzeng Zhang

Independently proposed the dual-slide adaptive gripper concept; conducted mechanism analysis and feasibility study, leading to long-term advising.

Remote Research Assistant – Shanghai AI Laboratory Apr 2026 – Present
Supervisor: Researcher Yuan Gao

Developing visual-servoing control algorithms for robotic-arm grasping and designing the mechanical end-effector.

Honors & Awards

National-Level Project (Principal Investigator), 2026 College Student Innovation and Entrepreneurship Training Program — "Zero-Lash Smart Grasping" 2026

Provincial-Level Project (Member), 2026 College Student Innovation Training Program — "Multi-Core Fiber Dynamic Optical Trap for Single-Cell Pathology" 2026

Crystal Prize for Scientific & Technological Innovation (top 10 across SZTU) 2025

Dean's Special Award for Rising Talent, Zhenxiong (Future Technology School, SZTU) 2024–2025

Outstanding Student Cadre, Shenzhen Technology University 2024, 2025

Excellence Prize, 15th "Challenge Cup" Student Entrepreneurship Competition (university round) 2025

Service & Leadership

- **Principal Investigator**, National-Level Project "Zero-Lash Smart Grasping", 2026 College Student Innovation and Entrepreneurship Training Program (2026.05 – present).
- **Project Member**, Provincial-Level Project on multi-core fiber dynamic optical trap for single-cell pathology, 2026 College Student Innovation Training Program (2026.05 – present).
- Conference reviewer for **IEEE/RSJ IROS 2026**.
- Head of Organization Department, Future Technology School, SZTU (2024.12 – present).
- Class Monitor & Youth League Secretary (2024.09 – 2025.10).
- Deputy Team Leader, 15th "Challenge Cup" Student Entrepreneurship Competition (2025).

Skills

Mechanism Design:	Parallel-jaw coupled grippers, linear linkages, parallelogram linkages, underactuated & tendon-driven hands; SolidWorks (modeling / assembly / kinematic simulation); Bambu Lab 3D printing; rapid prototyping and grasping experiment design.
Programming:	Python; basic experience with deep learning frameworks (point cloud / quantum circuit simulation).
Academic Writing:	IEEE journal & conference formatting; experienced with the full submission pipeline.
Languages:	Chinese (native); English (CET-4: 580; IELTS scheduled Dec 2026).